

Applied Machine Learning — Week 9

Interpretability & Explanations

Comprehensive Revision Notes

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What is Interpretability?

Definitions

“Interpretability is the degree to which a human can understand the **cause** of a decision.”
— Miller (2017)

“Interpretability is the degree to which a human can consistently **predict** the model’s result.”
— Kim et al. (2017)

When Do We Need Interpretability?

Why Interpretability Matters

- **Accountability:** if a model makes a mistake in a safety-critical application, we need to know *why* so it doesn’t happen again.
- **Debugging:** without interpretability, bias or discrimination may go undetected.
- **Social acceptance:** people will not accept unpredictable algorithms in their lives.
- Humans are interested in much more than just accuracy.

When Might We NOT Need Interpretability?

- Model errors have **low associated cost** — focus on accuracy is fine.
- Interpretability is **undesirable if it enables gaming** (e.g. automated loan decisions).
- Model has very high performance and failure modes are well understood.

Global vs. Local Understanding

Levels of Understanding

Global: “How does the model make predictions in general?”

- Can you comprehend the entire model at once? Hard even for linear models with many features.
- Can you understand how *parts* of the model affect predictions?

Local: “How did the model make *this specific* prediction?”

- Locally, a complex model may be well-approximated by something simpler.
- “For data similar to input x , features . . . are not important, and the prediction is roughly linear in . . .”
- Local explanations may be more accurate than global ones.

Two Approaches to Interpretability

The Two Approaches

1. **Intrinsic interpretability:** use a model that is inherently understandable.
2. **Post-hoc explanations:** given a black-box model that has made a prediction, produce an explanation of *why*.

Inherently Interpretable Models

Linear Models

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$$

Why interpretable: each coefficient β_d directly quantifies the effect of feature d on the output (holding other features constant). Sparse models (L1 / LASSO) are even more interpretable: only a few features have nonzero coefficients.

Limitation: can only capture linear relationships.

Decision Trees (but NOT Random Forests!)

Why interpretable: follow the path from root to leaf. Each split is an explicit, human-readable rule (“if age > 30 AND income < 50k, predict class 1”).

Limitation: individual trees have limited accuracy. Ensembles (RF, boosting) are far more accurate but sacrifice interpretability.

K-Nearest Neighbours

Why interpretable: “we predict $\hat{y}$ because you resemble these K other users.” Show the neighbours as the explanation.

Limitation: requires a meaningful distance metric; computationally expensive at test time.

Naïve Bayes

Naïve Bayes

A generative model:

$$p(y=k | x) = \frac{p(x | y=k) p(y=k)}{\sum_{j=1}^K p(x | y=j) p(y=j)}.$$

Why interpretable: the terms $p(x | y)$ represent “what does data look like given this class?” Naturally contrastive: compares “is this data more likely under class k or class k' ?”

Generalised Additive Models (GAMs)

GAMs

Extension of linear models with per-feature nonlinearities:

$$\hat{y} = \beta_0 + f_1(x_1) + f_2(x_2) + \dots,$$

where each f_d is a nonlinear function (e.g. a spline) estimated from data.

Why interpretable: holding other features constant, you can plot $f_d(x_d)$ to see exactly how feature d affects the prediction. No interaction terms (unless explicitly included as $f_{12}(x_1, x_2)$).

Exam Alert: Performance of Interpretable Models

Performance can be surprisingly good in many settings, especially with good feature engineering. Cynthia Rudin (2019): in a large-scale NYC electrical grid failure prediction project, the most accurate models were **sparse models with meaningful features** developed through iterative data exploration.

Post-Hoc Explanation Methods

Exam Alert: The “Models of Models” Problem

We had a phenomenon we didn’t understand, so we built a model. Now we have a model we don’t understand, so we build a model of the model. Have things gotten out of hand?

What Constitutes a Good Explanation?

Properties of Good Explanations

- **Contrastive**: explains why *this* prediction instead of *that* one (not just what’s salient).
- **Short / sparse**: focus on the most surprising or unusual aspects.
- **Faithful**: accurately reflects the actual decision-making process of the model.

Global Surrogates

Global Surrogate

1. Take your black-box model f and some inputs $x^{(i)}$.
2. Compute predictions $y^{(i)} = f(x^{(i)})$.
3. Fit an **interpretable model** to predict these predictions.

If the interpretable model has high accuracy at predicting the black-box’s predictions, it *may* be a useful explanation.

Problem: if the surrogate isn’t accurate, the “explanations” don’t reflect the black-box’s behaviour at all. Hard to know when to trust it.

LIME (Local Interpretable Model-Agnostic Explanations)

LIME

1. Take black-box model f and a specific input x_* .
2. Make **small perturbations** to x_* to produce “similar” inputs $x^{(i)}$.
3. Compute predictions $y^{(i)} = f(x^{(i)})$.
4. Fit an interpretable model (e.g. sparse linear model) to predict these predictions, **weighted by proximity to x_*** .

Unlike global surrogates, LIME approximates the model *locally*, so the fit is likely better.

Key problem: results depend heavily on the definition of “small perturbations,” which is data-dependent and difficult to select. Different perturbation strategies can give very different explanations.

Partial Dependence Plots (PDPs)

Partial Dependence

Choose feature d . For a model f and data $x^{(1)}, \dots, x^{(N)}$:

$$\hat{f}_d(x_d) = \frac{1}{N} \sum_{i=1}^N f(x_d, x_{\setminus d}^{(i)}).$$

This marginalises over all other features, showing the **average effect** of feature d on the prediction.

PDPs look like GAMs and are easy to interpret. However, they can be misleading in areas with little data, and they assume feature independence (ignore interactions).

Individual Conditional Expectation (ICE) Plots

Plot the per-individual curves $f(x_d, x_{\setminus d}^{(i)})$ for each i separately (instead of averaging). If lines cross, there are important **interactions** between feature d and other features.

Permutation Feature Importance

Permutation Feature Importance

1. Train the model and evaluate the loss L on test data.
 2. For each feature $d = 1, \dots, D$:
 - (a) Randomly shuffle (permute) the values of feature d across all test points.
 - (b) Evaluate the loss L_d on this permuted dataset.
 - (c) Feature importance: $F_d = L_d/L$ (ratio: how much worse is the model?).
 3. Sort features by descending importance.
- Scrambling “important” features will significantly increase the loss.

△ Correlated features confound permutation importance

If features d and d' are highly correlated, shuffling d breaks the correlation but the model can still predict well using d' . Feature d appears unimportant even though it is genuinely predictive. This is the same issue as LASSO zeroing correlated features.

Examples as Explanations

Counterfactual Explanations

Given input x with prediction $y = f(x)$, find the **closest** point x' such that $f(x') \neq f(x)$ (or $f(x') = y'$ for a desired alternative):

$$x' = \arg \min_{x'} d(x, x') \quad \text{subject to} \quad f(x') = y'.$$

Use a scaled L_1 norm for $d(x, x')$ to encourage changes in as few features as possible.

Interpretation: “Your loan was denied. If your income had been \$5,000 higher, it would have been approved.”

Pros: fidelity to the model is **guaranteed** (we actually query f); naturally contrastive.

Cons: optimisation is hard for images; multiple counterfactuals may exist.

Saliency Maps

Saliency Maps

For neural networks on images, identify which pixels have the largest effect on the output:

$$\text{Saliency} = \left| \frac{\partial f}{\partial x} \right| \quad (\text{magnitude of gradient w.r.t. input}).$$

Variants: gradient w.r.t. a hidden layer (GradCAM), integrated gradients, etc.

△ Saliency maps are unreliable!

Adebayo et al. (2018) showed that many saliency methods produce similar-looking maps even for **randomly initialised** networks — they may reflect *data structure* rather than *model reasoning*. Do not trust saliency maps uncritically.

The Rudin Argument: Do We Need Explanations?

Rudin (2019)

“**Stop explaining black-box machine learning models for high-stakes decisions** and use interpretable models instead.”

Key arguments:

- The trade-off between accuracy and interpretability is often **smaller than assumed**: with proper feature engineering, interpretable models can match black boxes.
- Explanations of black boxes have **no fidelity guarantee** — they might be completely wrong about why the model made a decision.
- For high-stakes applications (criminal justice, healthcare, lending), the risk of a wrong explanation is unacceptable.

Connections:

Interpretability, Fairness, and Privacy **Interpretability helps detect unfairness**: if you can see that a model relies on postcode (a proxy for race), you can intervene. Black-box models hide such dependencies.

Interpretability can conflict with privacy: a fully transparent model might reveal patterns about individuals in the training data.

All three topics (Weeks 9–10) form the “responsible ML” pillar of the course.

Summary of Methods

Method	Scope	Model-agnostic?	Faithful?	Key Weakness
Linear models	Global	N/A	Yes	Only linear effects
Decision trees	Global	N/A	Yes	Low accuracy
GAMs	Global	N/A	Yes	No interactions (unless explicit)
Global surrogate	Global	Yes	No guarantee	May not match black box
LIME	Local	Yes	No guarantee	Sensitive to perturbation choice
PDP / ICE	Global	Yes	Partial	Assumes feature independence
Permutation importance	Global	Yes	Partial	Correlated features confound
Counterfactual	Local	Yes	Yes	Optimisation hard for images
Saliency maps	Local	No (neural nets)	No guarantee	May reflect data, not model

Exam-Style Questions

Exam Q&A:

Q1 — LIME vs Global Surrogate (3 marks) **Q:** Compare LIME and global surrogates. Which is more likely to be faithful?

A: Both fit interpretable models to approximate a black box. Global surrogates approximate everywhere (poor fit likely, low fidelity). LIME approximates *locally* around a single prediction (better local fit). However, **neither** guarantees faithfulness — the surrogate might approximate well in terms of predictions but rely on completely different features than the black box. LIME is more likely to be locally faithful, but depends critically on the perturbation definition.

Exam Q&A:

Q2 — Permutation Importance with Correlation (3 marks) **Q:** Why might permutation feature importance be misleading with correlated features?

A: Shuffling feature d breaks its correlation with other features, creating unrealistic data combinations. But the model can still predict well using the correlated features d' , d'' , etc. So feature d appears unimportant (low F_d) even though it is genuinely predictive. The importance is “shared” among correlated features rather than attributed to each one.

Exam Q&A:

Q3 — Counterfactual Explanations (3 marks) **Q:** What advantage do counterfactual explanations have over saliency maps?

A: Counterfactuals are **faithful by construction**: they query the actual model f to find x' with $f(x') \neq f(x)$, so the explanation directly reflects the model’s behaviour. Saliency maps compute gradients that may not reflect the model’s actual reasoning (Adebayo et al. 2018 showed they can look similar even for random networks). Counterfactuals are also naturally **contrastive** and **actionable** (“change feature d by this much”).

Exam Q&A:

Q4 — When to Use Interpretable Models (2 marks) **Q:** Give one argument for and one against always using interpretable models.

A: For: Rudin (2019) shows that with good feature engineering, interpretable models often match black boxes in accuracy, while providing guaranteed faithful explanations — essential for high-stakes decisions. **Against:** for complex inputs like images, video, or raw audio, interpretable models (linear, trees) simply cannot achieve competitive accuracy. Deep learning is necessary, and post-hoc explanations are the only option.

Exam Q&A:

Q5 — GAMs (2 marks) **Q:** Why are GAMs more interpretable than a general neural network, despite both being nonlinear?

A: GAMs are *additive*: $\hat{y} = \beta_0 + \sum_d f_d(x_d)$. Each f_d can be plotted independently, showing exactly how feature d affects the output. There are no interaction terms (unless explicitly modelled). A neural network has arbitrary nonlinear interactions between all features, making it impossible to isolate individual feature effects.

End of Week 9 Revision Notes.

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